RAS vs Claude Life Sciences: the same research, a fraction of the compute

We ran 11 life-sciences research tasks — variant lookups, paper extraction and paper-versus-database cross-checks — through Claude Life Sciences and Adya RAS, and measured the tokens consumed and the time to a finished answer.

−74%
Input tokens
21.4M → 5.5M across all runs

−87%
Output tokens
323K → 42K across all runs

3.9×
Fewer total tokens
21.7M → 5.6M, a 74% reduction

2.9×
Faster to answer
89 min → 31 min of total run time

What we tested

Seven prompts across three families of increasing difficulty. Claude Life Sciences ran on Sonnet 5.5 throughout; RAS ran four tasks on Haiku 4.5 and seven on Sonnet 4.6.

A · Database lookups
Single-variant questions answered from live sources: ClinVar, gnomAD v4 and CPIC.

B · Paper extraction
Grounded answers drawn from supplied papers, with the section cited.

C · Cross-checks
Claims in a paper checked against ClinGen, ClinVar, gnomAD and CIViC, with differences flagged.

At a glance

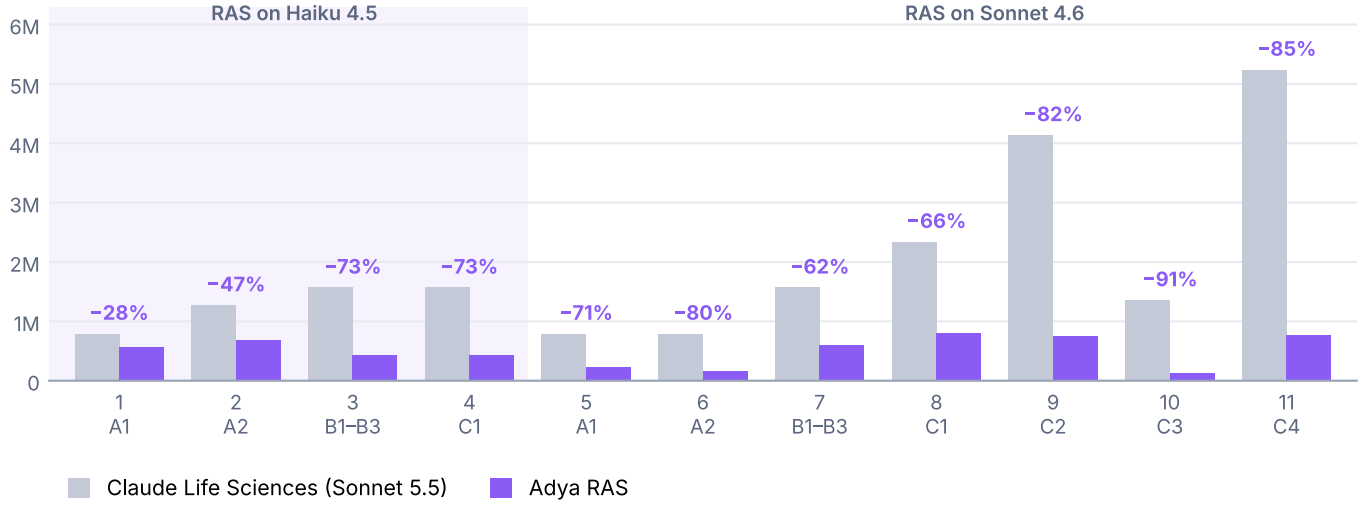
METRIC (ALL 11 RUNS)	CLAUDE LIFE SCIENCES	ADYA RAS	REDUCTION
Input tokens	21,397,707	5,510,752	−74.2%
Output tokens	322,865	42,248	−86.9%
Total tokens	21,720,572	5,553,000	−74.4%
Time taken	5,342s (89 min)	1,841s (31 min)	−65.5%

Token consumption

Each pair of bars is one run. The figure above each pair is RAS's reduction against Claude Life Sciences. Shaded runs used Haiku 4.5 on the RAS side.

Input tokens

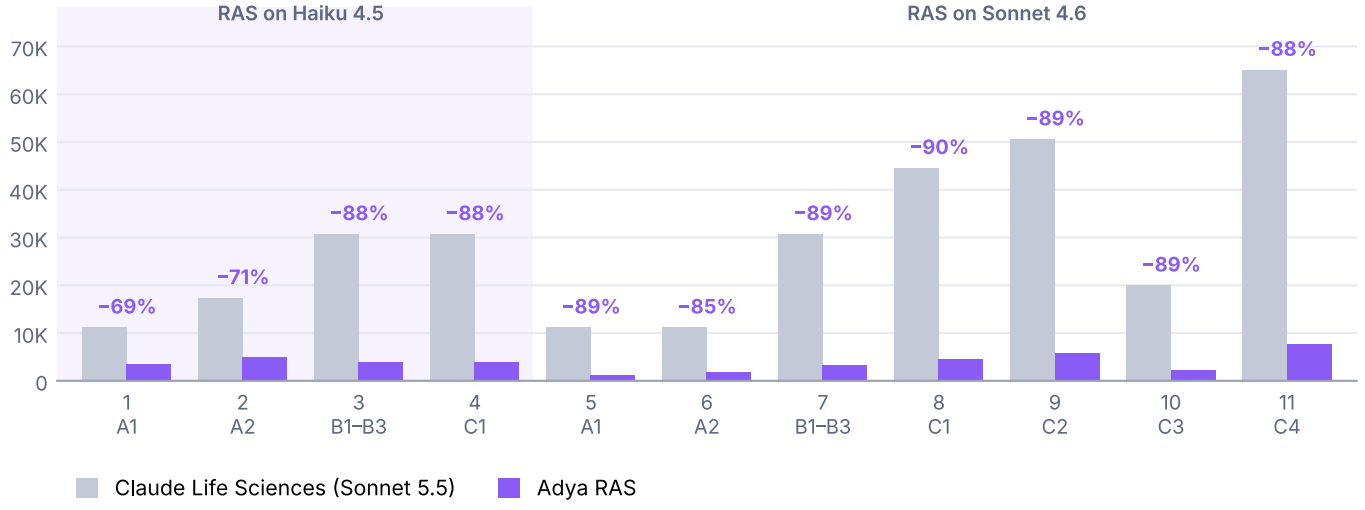
Context read per run — sources, documents and tool results



RAS read **74% less context** overall. The gap widens on the hardest tasks: on C4 (CIViC evidence for paper 05) Claude Life Sciences read 5.24M tokens against RAS's 0.77M.

Output tokens

Tokens written per run



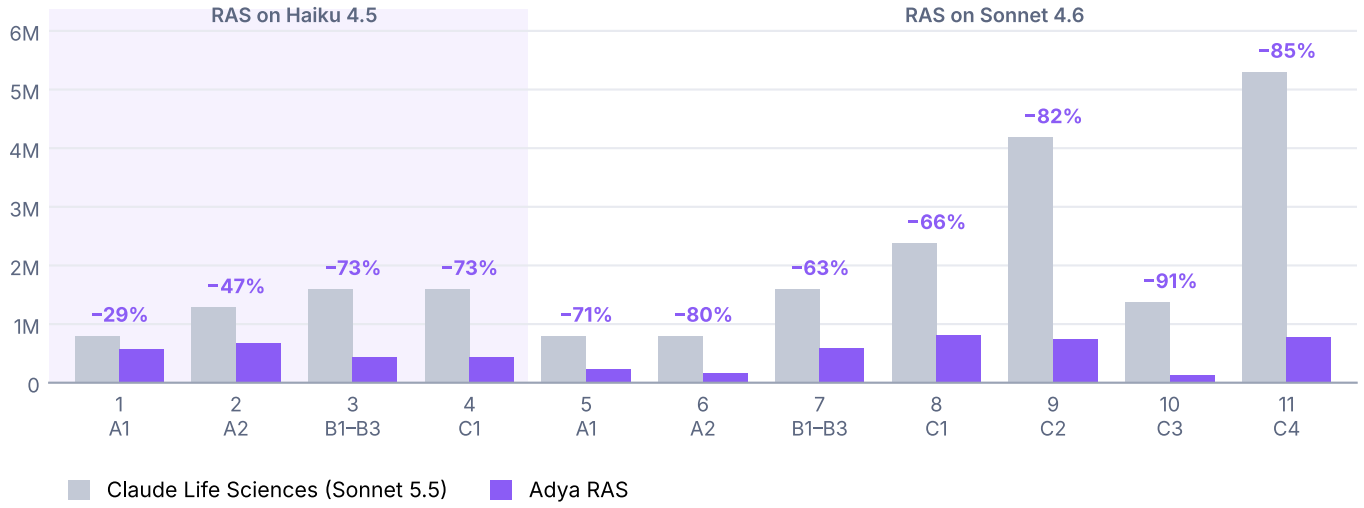
RAS wrote **7.6× fewer output tokens**, cutting 69–90% on every run — less intermediate reasoning to pay for and a tighter deliverable to review.

Total cost and speed

Total tokens is the main driver of cost per question; time taken is how long an analyst waits for the answer.

Total tokens

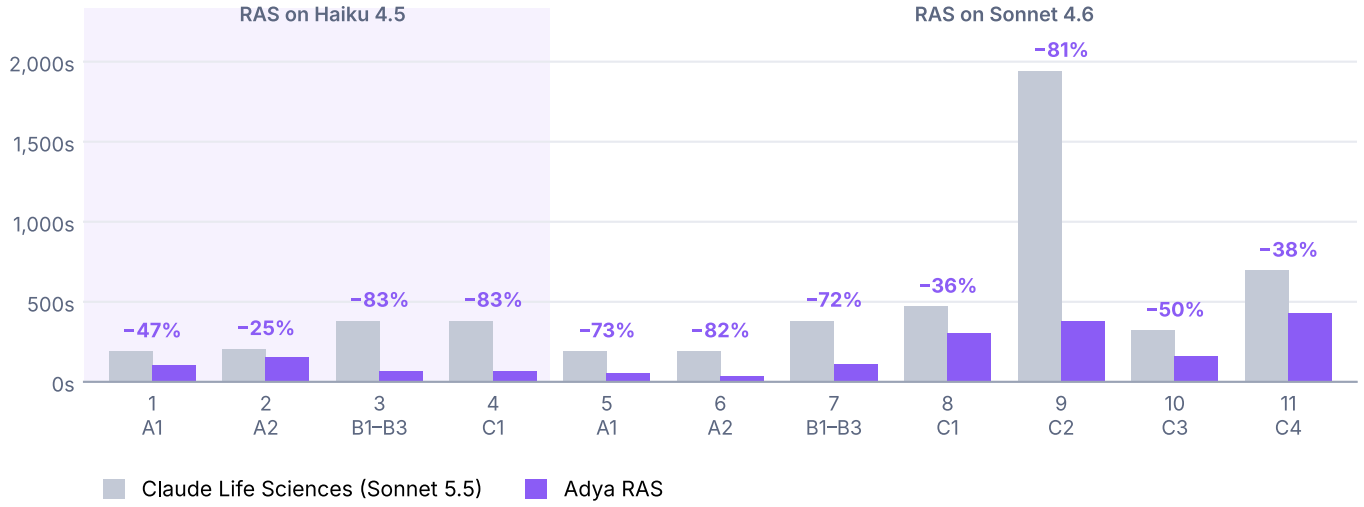
Input plus output tokens per run



Across the 11 runs RAS used **5.6M tokens against 21.7M**. Savings were 29–73% on Haiku 4.5 and 63–91% on Sonnet 4.6, where C3 fell from 1.37M tokens to 0.13M.

Time taken

Seconds from question to finished answer



RAS finished every run sooner, **2.9× faster in aggregate**. The ACMG/AMP classification task (C2) dropped from 32 minutes to just over 6.

Run-by-run results

Total tokens and time for every run, with RAS's reduction against Claude Life Sciences.

#	TASK	QUESTION	RAS MODEL	CLAUDE TOKENS	RAS TOKENS	CLAUDE TIME	RAS TIME	TOKEN CUT	TIME CUT
1	**A1**	ClinVar classification of ATP7B p.His1069Gln — review status, submitters, conflicts	Haiku 4.5	800K	569K	190s	101s	−29%	−47%
2	**A2**	gnomAD v4 global and South Asian frequencies for CYP2C19 *2, *3, *17, plus CPIC clopidogrel guidance	Haiku 4.5	1.29M	679K	204s	153s	−47%	−25%
3	**B1–B3**	Three extraction questions across papers 01, 02 and 04 (HCM genes, MYBPC3 D389V evidence, Indian PGx)	Haiku 4.5	1.60M	430K	381s	63s	−73%	−83%
4	**C1**	Every gene in paper 03 (DCM) checked against current ClinGen classification and inheritance	Haiku 4.5	1.60M	430K	381s	63s	−73%	−83%
5	**A1**	ClinVar classification of ATP7B p.His1069Gln — review status, submitters, conflicts	Sonnet 4.6	800K	234K	190s	51s	−71%	−73%
6	**A2**	gnomAD v4 global and South Asian frequencies for CYP2C19 *2, *3, *17, plus CPIC clopidogrel guidance	Sonnet 4.6	800K	156K	190s	35s	−80%	−82%
7	**B1–B3**	Three extraction questions across papers 01, 02 and 04 (HCM genes, MYBPC3 D389V evidence, Indian PGx)	Sonnet 4.6	1.60M	595K	381s	108s	−63%	−72%
8	**C1**	Every gene in paper 03 (DCM) checked against current ClinGen classification and inheritance	Sonnet 4.6	2.38M	810K	469s	300s	−66%	−36%
9	**C2**	MYBPC3 D389V — ClinVar, gnomAD South Asian frequency, ACMG/AMP criteria, suggested classification	Sonnet 4.6	4.18M	749K	1,942s	377s	−82%	−81%
10	**C3**	Paper 04 CYP2C19 / CYP2C9 frequencies vs gnomAD v4 South Asian; flag gaps above 5 points	Sonnet 4.6	1.37M	129K	320s	161s	−91%	−50%
11	**C4**	Current CIViC PARP-inhibitor evidence for the HRR genes in paper 05, plus newer publications	Sonnet 4.6	5.30M	773K	694s	429s	−85%	−38%

The harder the task, the bigger the gap: on database lookups (family A) RAS cut total tokens by 56%; on paper-versus-database cross-checks (family C) the cut rose to 81%.

Method: each task was run once per system on the same prompt. Claude Life Sciences used Sonnet 5.5; RAS used Haiku 4.5 (runs 1–4) and Sonnet 4.6 (runs 5–11). This benchmark measures efficiency — tokens and time — and does not score answer quality.

Same life-sciences questions. ~4× fewer tokens. ~3× faster answers.